

Mean, Median, and Baselines

Project: Responsible Plant Classifier

1. Your mission

Your learning target

I can fit mean and median constant baselines on training targets.

I can compare constant predictions on separate records.

I can keep fitted constants independent of evaluation answers.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

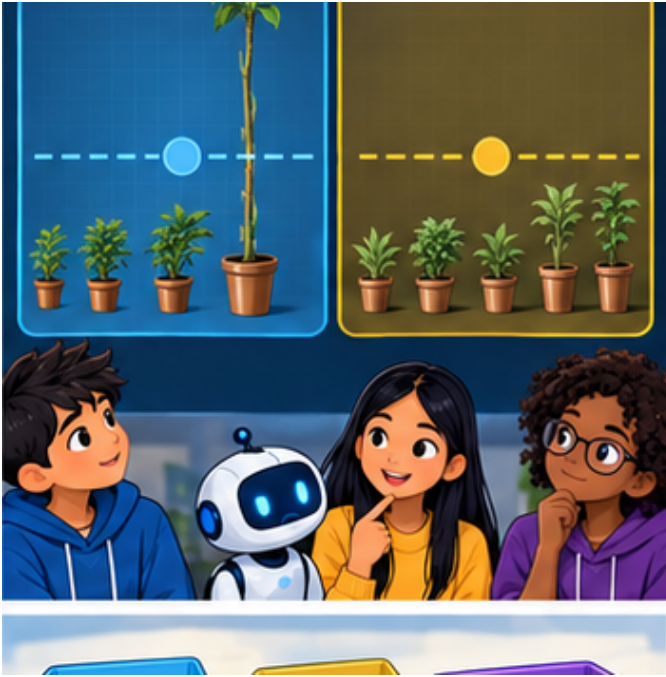
Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Mean and median are transparent regression baselines; outliers affect them differently.



LEARNING TARGET For training targets [1,3,11], compute mean and median constant baselines. Compare total absolute training error for each. Where must final performance be measured?

Builder name and experiment date:

Readiness for the connected explanation

Why can signed residuals hide error size?

This week's end goal

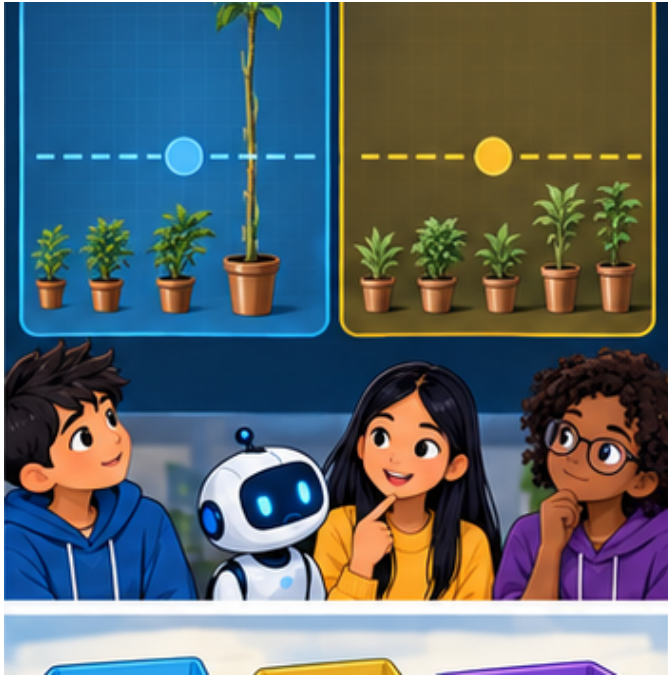
By the end, I can explain mean, median, and baselines using a traced example and a stated limitation.

2. Notice the evidence

PICTURE ENTRY

Notice → **Predict** → **Explain**

Begin with visible evidence; do not explain more than the picture and information support.



MYSTERY The team must use mean, median, and baselines to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

A closer explanation

Before a complex regression model, fit a simple constant predictor. A mean baseline uses the training average. A median baseline uses the middle training target. They ignore features but provide a comparison that is easy to reproduce. Evaluation targets must not determine the constants.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines constant prediction baselines.

3. See how it works

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.



IN

INPUT

Choose and justify a baseline under an extreme value.



ML

MODEL

Mean and median are transparent regression baselines; outliers affect them differently.



OUT

EVIDENCE

Choose and justify a baseline under an extreme value.

QUESTION Choose and justify a baseline under an extreme value.

MODEL IDEA Mean and median are transparent regression baselines; outliers affect them differently.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Heights are 10, 11, 12, 13, and 54 cm. Mean = $100 \div 5 = 20$ cm.

Draw a fresh data → model → prediction → evaluation trace:

A closer explanation

Training targets [2, 4, 6, 12] have mean $24/4=6$ and median $(4+6)/2=5$. On separate targets [4, 8], the mean baseline predicts [6, 6] with absolute errors [2, 2], averaging 2. The median baseline predicts [5, 5] with errors [1, 3], also averaging 2.

```
from statistics import mean, median

training = [2, 4, 6, 12]
validation = [4, 8]

for name, value in [("mean", mean(training)),
                    ("median", median(training))]:
    mae = mean(abs(y - value) for y in validation)
    print(name, value, mae)

# Both validation MAEs are 2; constants differ.
```

Trace the calculation on paper before running this optional Python check. Use the stated fictional inputs and compare intermediate values. The code checks this example; your independent workbook case uses

different evidence.

Connect the picture and the explanation

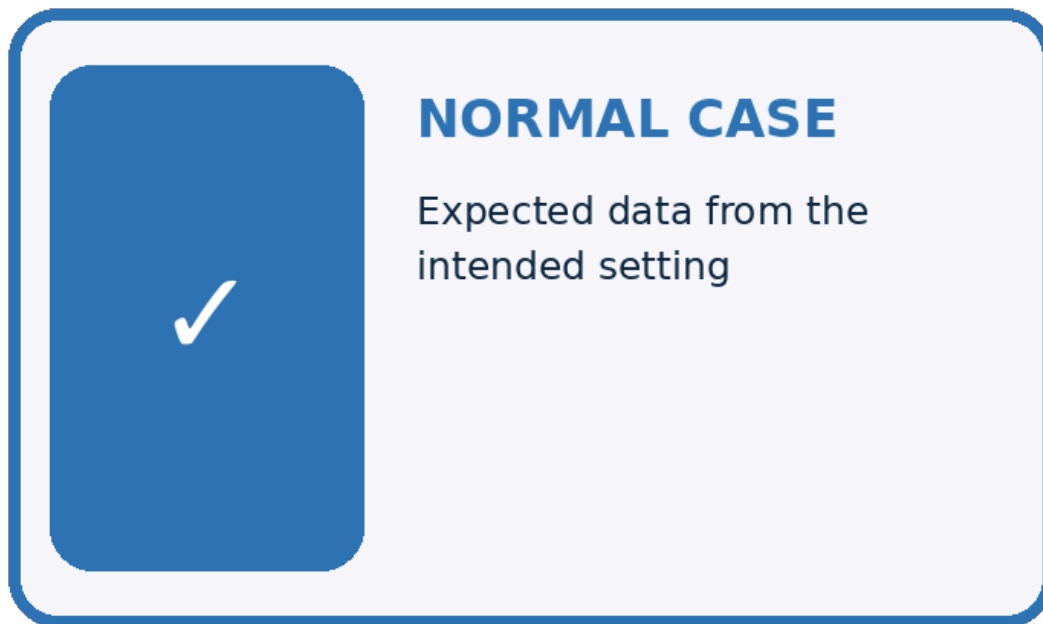
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

EVALUATION DESIGN

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





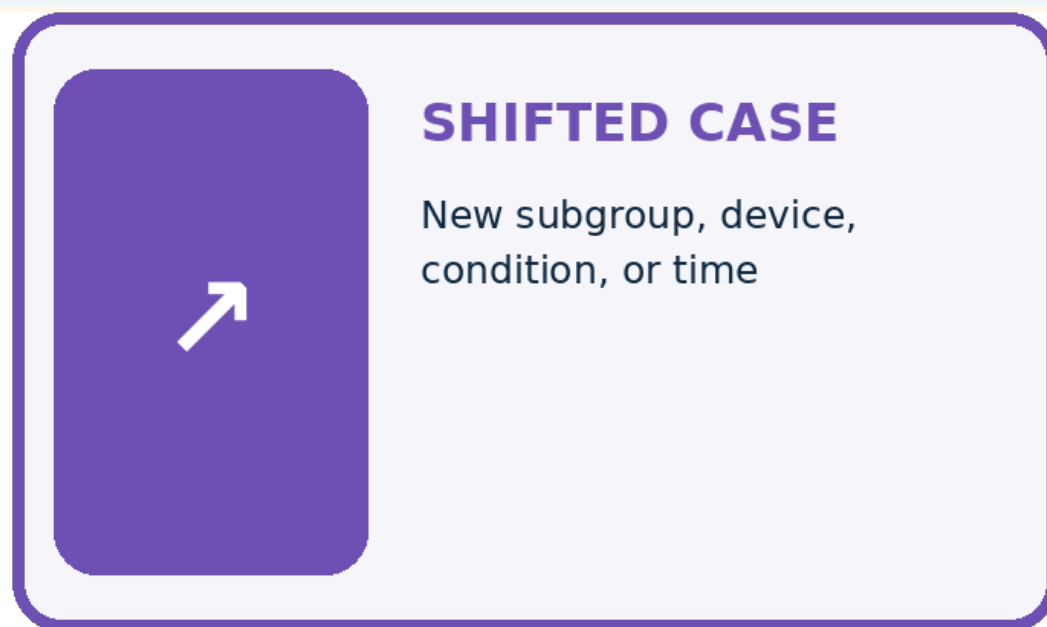
BOUNDARY CASE

Value near a threshold or class boundary



MISSING CASE

Unknown field, label, or necessary evidence



NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change: Replace training 20 with 8 while keeping evaluation unchanged.

Prediction: Both fitted constants become 5 cm and both evaluation MAEs remain 3 cm.

Reason: Mean is now $20/4=5$; median stays 5. The earlier training outlier affected the mean.

5. Words that unlock the idea

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

MEAN

sum divided by count

2

MEDIAN

middle ordered value

3

OUTLIER

value far from most observations

4

LOCATION BASELINE

single typical value predicted for all cases

Term · Precise meaning · My ML example

Term: mean

Precise meaning: sum divided by count

My ML example: _____

Term: median

Precise meaning: middle ordered value

My ML example: _____

Term: outlier

Precise meaning: value far from most observations

My ML example: _____

Term: location baseline

Precise meaning: single typical value predicted for all cases

My ML example: _____

MISCONCEPTION TO REPAIR The mean is always the fairest and most scientific summary.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word · Meaning

Word: Constant baseline

Meaning: One fixed prediction for every query

Word: Training mean

Meaning: Sum of training targets divided by their count

Word: Training median

Meaning: Middle value of sorted training targets

Word: Absolute error

Meaning: Magnitude of observed minus predicted

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

WEEK 20 • WORKED EVIDENCE

- 1 Heights are 10, 11, 12, 13, and 54 cm.
- 2 Mean = $100 \div 5 = 20$ cm.
- 3 Median is 12 cm, the middle ordered value.
- 4 For typical-height absolute error, median is more robust; still report why 54 is present.

STEP 1 Heights are 10, 11, 12, 13, and 54 cm.

STEP 2 Mean = $100 \div 5 = 20$ cm.

STEP 3 Median is 12 cm, the middle ordered value.

STEP 4 For typical-height absolute error, median is more robust; still report why 54 is present.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

A closer explanation

Constant prediction baselines at a glance

Part · Evidence or action · Interpretation

Part: Training targets

Evidence or action: Fit mean 6

Interpretation: Freeze constant

Part: Training targets

Evidence or action: Fit median 5

Interpretation: Freeze constant

Part: Evaluation targets

Evidence or action: Compare fixed predictions

Interpretation: Both MAE 2 here

Mean and median answer different training objectives. A mean minimizes total squared error among constants; a median minimizes total absolute error. This does not guarantee either wins on a new evaluation set. With extreme values, the mean can move much more than the median.

Calculate MAE for both baselines on [4, 8]. Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Mean, Median, and Baselines

Fit two constant height baselines and compare them on held-out examples.

The facts we will use

Training heights [2, 4, 6, 20] cm. Evaluation heights [4, 10] cm. Fit constants only from training.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Mean, Median, and Baselines

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Fit mean

Training mean= $(2+4+6+20)/4=8$ cm.



2. Fit median

Training median= $(4+6)/2=5$ cm.



3. Evaluate constants

Mean MAE= $(4+2)/2=3$ cm; median MAE= $(1+5)/2=3$ cm.

CHANGE ONE THING

Replace training 20 with 8 while keeping evaluation unchanged.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Fit mean

What happens and why: Training mean= $(2+4+6+20)/4=8$ cm.

Step: 2. Fit median

What happens and why: Training median= $(4+6)/2=5$ cm.

Step: 3. Evaluate constants

What happens and why: Mean $MAE=(4+2)/2=3$ cm; median $MAE=(1+5)/2=3$ cm.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Mean $MAE=(4+2)/2=3$ cm; median $MAE=(1+5)/2=3$ cm. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Mean and median have different loss properties, but their held-out performance depends on the particular evaluation cases.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



COMPARE • DIAGNOSE • REPAIR

Tempting claim: The mean is always the fairest and most scientific summary. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: passes

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: defined behavior

Actual: _____

Likely cause / next test: _____

Evidence record: missing case

Expected: safe fallback

Actual: _____

Likely cause / next test: _____

Evidence record: shifted case

Expected: measured limit

Actual: _____

Likely cause / next test: _____

CLAIM TO CHALLENGE The mean is always the fairest and most scientific summary.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

For an even number of values, this course averages the two middle values to report one median. Keep a frozen baseline across comparisons and show its evaluation score. A model that fails to beat it may still reveal a data or metric issue worth investigating.

Why not predict the evaluation mean as the baseline?

8. Practise together

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



FREEZE THE EXPERIMENT

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Replacing training 12 by 40 gives mean 13 and median 5. That change is a data change to record and investigate, not an instruction to delete an inconvenient observation. Decide whether the extreme value is valid using the measurement policy and provenance.

Practise with less help: Score separately

Use workbook W2 for the connected example “Constant prediction baselines”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Constant prediction baselines. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Baseline Tournament: compute mean and median, predict every case, compare MAE, and explain the outlier's effect.

1. Write the hypothesis and expected evidence before running.
2. Baseline Tournament: compute mean and median, predict every case, compare MAE, and explain the outlier's effect.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

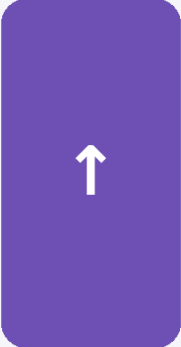
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



ML BUILDER MASTERY
1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

DEFINE Use all four terms precisely: mean, median, outlier, location baseline.

CALCULATE Reproduce the worked evidence with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE The mean is always the fairest and most scientific summary.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

11. Connect your project

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.

**RESPONSIBLE PLANT CLASSIFIER**

Document any numeric baseline used in Plant Classifier feature checks, including unit, distribution, and outlier decision.

THIS WEEK'S CHECKPOINT Document any numeric baseline used in Plant Classifier feature checks, including unit, distribution, and outlier decision.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



INDEPENDENT ML BUILDER PROOF
Choose and justify a baseline under an extreme value.

REQUIRED PROOF Choose and justify a baseline under an extreme value.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Absolute Error and MAE.

13. Study a complete case

Week 20 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Training heights are 2,4,12 cm. Compare constant mean and median baselines.

The question

Compute both and training absolute errors.

Worked explanation

Mean=6, median=4. Mean errors 4,2,6 total12; median errors 2,0,8 total10. Median minimises absolute error here.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Mean and median baselines use the same observations

Evidence or stage · Value or result

Evidence or stage: Observation A

Value or result: 2

Evidence or stage: Observation B

Value or result: 4

Evidence or stage: Observation C

Value or result: 12

Evidence or stage: Mean baseline

Value or result: 6

Evidence or stage: Median baseline

Value or result: 4

Training heights are 2,4,12 cm. Compare constant mean and median baselines.

Compute both and training absolute errors.

Mean=6, median=4. Mean errors 4,2,6 total12; median errors 2,0,8 total10. Median minimises absolute error here.

14. Try a new case independently

Independent case: Apply independently

Training [1, 3, 5, 15], evaluation [3, 9]. Fit both constants and calculate MAE.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 20 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

For training heights 2, 3 and 10 cm, calculate mean and median. Which constant is less influenced by the high value?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Document any numeric baseline used in Plant Classifier feature checks, including unit, distribution, and outlier decision.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

For training targets [1,3,11], compute mean and median constant baselines. Compare total absolute training error for each. Where must final performance be measured?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.